Phase 3: Tokenizer Analysis

Status: Final 50,000-token tokenizer (50,000 vocabulary entries). **Tokenizer:** `models/tokenizer.json`.

Training Runtime

Full-corpus training used 337.1s for counting and 173.7s for merge learning (510.8s total). Recorded system: Linux-7.2.9-200.fc44.x86_64-x86_64-with-glibc2.43; Python 3.14.8; 12 CPU cores; peak memory 2332 MB. CPU model: not reported by platform.processor().

1. Merge Analysis

Merge	Input IDs	Bytes	Valid UTF-8 alone	Decoded token
1	216 + 167	2	yes	"ا"
2	219 + 140	2	yes	"ی"
3	32 + 216	2	no	invalid UTF-8: b' '
4	218 + 169	2	yes	"ک"
5	32 + 217	2	no	invalid UTF-8: b' '
6	216 + 177	2	yes	"ر"
7	217 + 136	2	yes	"و"
8	219 + 129	2	yes	"ہ"
9	219 + 146	2	yes	"ے"
10	32 + 260	3	yes	" ک"
11	217 + 134	2	yes	"ن"
12	257 + 216	3	no	invalid UTF-8: b"
13	217 + 132	2	yes	"ل"
14	217 + 133	2	yes	"م"
15	261 + 133	3	yes	" م"
16	218 + 186	2	yes	"ں"
17	32 + 257	3	yes	" ا"
18	216 + 170	2	yes	"ت"
19	32 + 264	3	yes	" ہ"
20	258 + 272	4	yes	"یں"
21	216 + 179	2	yes	"س"
22	216 + 175	2	yes	"د"
23	13 + 10	2	yes	""
24	259 + 179	3	yes	" س"
25	218 + 190	2	yes	"ھ"
26	259 + 168	3	yes	" ب"
27	218 + 175	2	yes	"گ"
28	266 + 258	5	yes	" کی"
29	266 + 265	5	yes	" کے"
30	216 + 168	2	yes	"ب"
31	261 + 190	3	yes	" پ"
32	257 + 267	4	yes	"ان"
33	261 + 134	3	yes	" ن"
34	263 + 262	4	yes	"ور"

Merge	Input IDs	Bytes	Valid UTF-8 alone	Decoded token
35	259 + 170	3	yes	" ت"
36	219 + 148	2	yes	"ِ"
37	259 + 172	3	yes	" ج"
38	257 + 262	4	yes	"ار"
39	32 + 268	4	no	invalid UTF-8: b' '
40	271 + 276	7	yes	" میں"
41	259 + 175	3	yes	" د"
42	217 + 129	2	yes	"ف"
43	217 + 130	2	yes	"ق"
44	217 + 185	2	yes	"ٹ"
45	261 + 136	3	yes	" و"
46	216 + 185	2	yes	"ع"
47	257 + 269	4	yes	"ال"
48	275 + 265	5	yes	" ہے"
49	268 + 166	4	yes	"ائ"
50	259 + 177	3	yes	" ر"
5,000	1710 + 4736	15	yes	" پریشانی"
10,000	301 + 8576	9	yes	" ورحم"
15,000	338 + 14363	13	yes	" انوکھی"
20,000	645 + 392	11	yes	" کاملہ"
25,000	388 + 277	5	yes	" ٹس"
30,000	854 + 378	8	yes	"منون"
35,000	4606 + 688	11	yes	" چکانا"
40,000	500 + 35410	9	yes	" میَں"
45,000	13341 + 115	4	yes	"atis"
49,743	480 + 308	5	yes	" غپ"

Two-byte Urdu letters rebuilt in the first 50: 1: "ا", 2: "ی", 4: " ک", 6: "ر", 7: "و", 8: "ہ", 9: "ے", 11: "ن", 13: "ل", 14: "م", 16: "ں", 18: "ت", 21: "س", 22: "د", 25: "ھ", 27: " گ", 30: "ب", 42: "ف", 43: "ق", 44: "ٹ", 46: " ع" Early merges invalid as standalone UTF-8 (first 50): 3, 5, 12, 39 Letter-forming merges rebuild UTF-8 Urdu characters; merges 3, 5, 12, 39 are partial byte sequences and cannot be decoded alone. Merge 20 first forms the multi-letter corpus word "یں".

20 Longest Learned Tokens

Token ID	UTF-8 bytes	Token
26755	32	"----------"
35817	30	"چندرابابونائیڈو"
19038	29	" انسائیکلوپیڈیا"
36509	29	" انٹر کانٹینینٹل"
29505	28	"ٹارتھیمپٹنشائر"
10120	27	" انادکارپوریٹڈ"
18324	27	" خیبرپختونخواہ"
31708	27	" پارلیمنٹیرینز"
38710	27	" انگیزانکشافات"
42258	27	" لاہورہائیکورٹ"
49866	27	" غیرجانبدارانہ"

Token ID	UTF-8 bytes	Token
26680	26	"——————————."
37511	26	"خیبر پختونخواہ"
47785	26	"لاہور ہائیکورٹ"
7682	25	"خیبر پختونخوا"
23648	25	"امیر المومنین"
26325	25	"امیر المؤمنین"
27489	25	"اورنامور مصنف"
32915	25	"دارالحکومتوں"
37547	25	"انفرااسٹرکچر"

2. Tokens per Word

Word counts use whitespace-delimited words; end-of-text markers are excluded.

Input	Words	Urdu BPE tokens/word	GPT-2 tokens/word	o200k_base tokens/word
test.txt	2,367,063	1.222	6.102	1.638
News	15	1.133	5.733	1.533
Poetry	15	1.133	6.133	1.533
Urdu with English	12	1.500	4.000	1.333
Roman Urdu	9	3.000	2.667	2.000
Numbers and dates	11	2.455	6.091	2.455
Diacritics	7	2.429	7.857	3.857
Personal names	8	1.500	7.750	2.625

On test.txt, Urdu BPE used 1.222 tokens per word, versus 6.102 for GPT-2 and 1.638 for o200k_base. Roman Urdu, rare spellings, and mixed-script inputs remain useful stress cases.

3. Inputs Handled Poorly

The three selected inputs with the highest Urdu-BPE token/word ratio are:

Input type	Urdu BPE tokens/word	Example	Interpretation
Roman Urdu	3.000	“Aaj mausam bohat khubsurat hai, chalo Islamabad chalte hain.”	Romanized spellings vary and do not share the Urdu-script merges learned by this tokenizer.
Numbers and dates	2.455	"قیمت ۱۲۵۰ روپے ہے؛ تاریخ 11-10-2026 اور وقت 9:45 بجے ہے۔"	Digits and punctuation form combinations that are not reusable word-like units.
Diacritics	2.429	“اَللّٰہُ اَکْبَرْ، اُردُو کے اعراب اور حرکات”	Combining marks add UTF-8 bytes and are relatively sparse, so words may remain split.

Roman Urdu was hardest at 3.000 tokens per word. The examples are diagnostic rather than a representative evaluation set.